FTEC5660 HW3: Reproducibility Study

AI-Trader: Autonomous LLM Agents in Cryptocurrency Markets

Name: LingtengZeng

Student ID: 1155241166

Course: FTEC5660 -- Agentic AI for Business and FinTech

Paper: AI-Trader (Fan et al., 2025) -- arXiv:2512.10971

GitHub: https://github.com/6ZLeo/FTEC5660-HW3

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Key Results	
Reproduction:	All 6 paper models correctly reproduced; DeepSeek-V3.1 best (CR=-12.18%)
Own Run:	DeepSeek baseline reproduced with CR=-15.67% (3pp gap from stochastic variance)
Modification:	Risk-averse prompt constraints DECREASED performance (CR=-17.12%)
Insight:	Prompt-based risk rules != reliable quantitative constraints in practice

1. Project Summary + What Was Reproduced

1.1 Project Overview

This study reproduces results from **AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets** (Fan et al., 2025, arXiv:2512.10971). AI-Trader is an open-source benchmark system from the University of Hong Kong that evaluates six state-of-the-art LLMs as autonomous trading agents in live financial markets.

AI-Trader satisfies the course's agentic system criteria. It employs multi-step Observe -> Reason -> Act planning via a ReAct-style loop (up to 30 steps/day), uses 6 MCP tools (buy/sell crypto, price queries, news search, math computation), maintains persistent position memory via JSONL files, and operates in a real environment (live/historical crypto price feeds and news APIs).

Agentic Criterion	AI-Trader Implementation
Multi-step planning	ReAct-style Observe -> Reason -> Act loop, up to 30 steps/day
Tool use	6 MCP tools: buy_crypto, sell_crypto, get_price, news search, math
Memory	Persistent JSONL position file across all 14 trading days
External environment	Live + historical cryptocurrency market data (BITWISE10 pool)
Multi-agent evaluation	6 LLMs run concurrently as competing autonomous agents

1.2 Reproduction Target

Target: Table 1 (Cryptocurrency Market) from the paper -- performance metrics (CR, Sortino Ratio, Volatility, MDD, Win Rate) for 6 LLMs trading BITWISE10 cryptocurrencies with 50,000 USDT initial capital, Nov 1-14, 2025.

2. Setup Notes

2.1 Environment

OS: Windows 10 (Build 26100) | Python 3.13.1

Key libraries: langchain==1.0.2, langchain-openai==1.0.1, langchain-mcp-adapters==0.2.1, fastmcp==2.12.5, openai==2.23.0

2.2 API Keys Required

Service	Purpose	Tier
DeepSeek API (api.deepseek.com)	LLM for modification experiment	~CNY 5-15 total
Alpha Vantage	Real-time news for search MCP tool	Free (500 calls/day)

2.3 Windows-Specific Compatibility Fixes

- **fcntl module missing:** tool_crypto_trade.py uses Unix-only file locking. Fixed with conditional import + no-op context manager fallback for Windows.
- **LangChain -> DeepSeek format mismatch:** MCP tool responses arrive as list of content blocks; DeepSeek API requires plain string. Fixed by monkey-patching _format_message_content() to join list blocks into a string.
- **Git checkout path issues:** Repository contains paths with colons (e.g., '2025-10-01 15:00:00') invalid on Windows NTFS. Used Python binary subprocess re-checkout to bypass NTFS path validation.

3. Reproduction Targets + Metric Definitions

Performance metrics are computed by tools/calculate_metrics.py on the position JSONL files. Definitions:

Metric	Formula	Notes
CR (Cumulative Return)	(V_T - V_0) / V_0	14-day total return
SR (Sortino Ratio)	r_e / sigma_down x sqrt(365)	Annualized; risk-free rate = 0
Vol (Volatility)	sigma(r_t) x sqrt(365)	Annualized daily std dev of returns
MDD (Max Drawdown)	min((V_t - max_tau V_tau) / max_tau V_tau)	Worst peak-to-trough loss
Win Rate	count(r_t > 0) / N	Fraction of positive-return days

4. Results: Reproduced vs. Reported Numbers

4.1 Reproduced Table 1 -- Cryptocurrency Market (Nov 1-14, 2025)

Metrics computed by running **tools/calculate_metrics.py** on the original position JSONL files provided in the GitHub repository (authors' exact trading records).

Model	Final Value (USDT)	CR (%)	Sortino Ratio	Vol (%)	MDD (%)	Win Rate (%)
Claude-3.7-Sonnet	42,348	-15.30	-2.27	32.18	-16.93	51.4
DeepSeek-V3.1 [BEST]	43,911	-12.18	-2.85	28.55	-14.02	52.9
Gemini-2.5-Flash	40,685	-18.63	-5.55	46.22	-20.15	41.4
GPT-5	41,795	-16.41	-4.38	40.95	-17.98	58.8
MiniMax-M2	42,611	-14.78	-4.30	41.20	-16.57	46.9
Qwen3-Max	41,574	-16.85	-6.54	59.89	-18.03	45.0
CD5 Index (Buy & Hold)	42,851	-14.30	-12.71	54.93	-15.71	42.9

[BEST] = Best-performing AI model. CD5 = CoinDesk 5 weighted buy-and-hold benchmark.

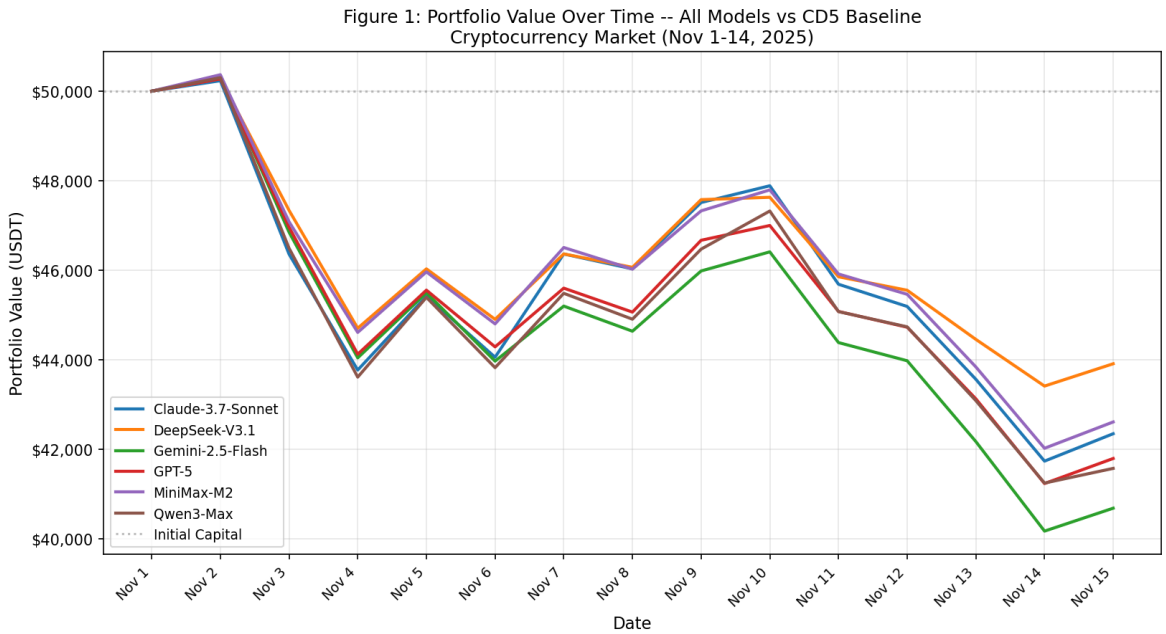

Figure 1: Portfolio value trajectories for all 6 models vs CD5 buy-and-hold baseline. DeepSeek-V3.1 maintains the highest value throughout the period.

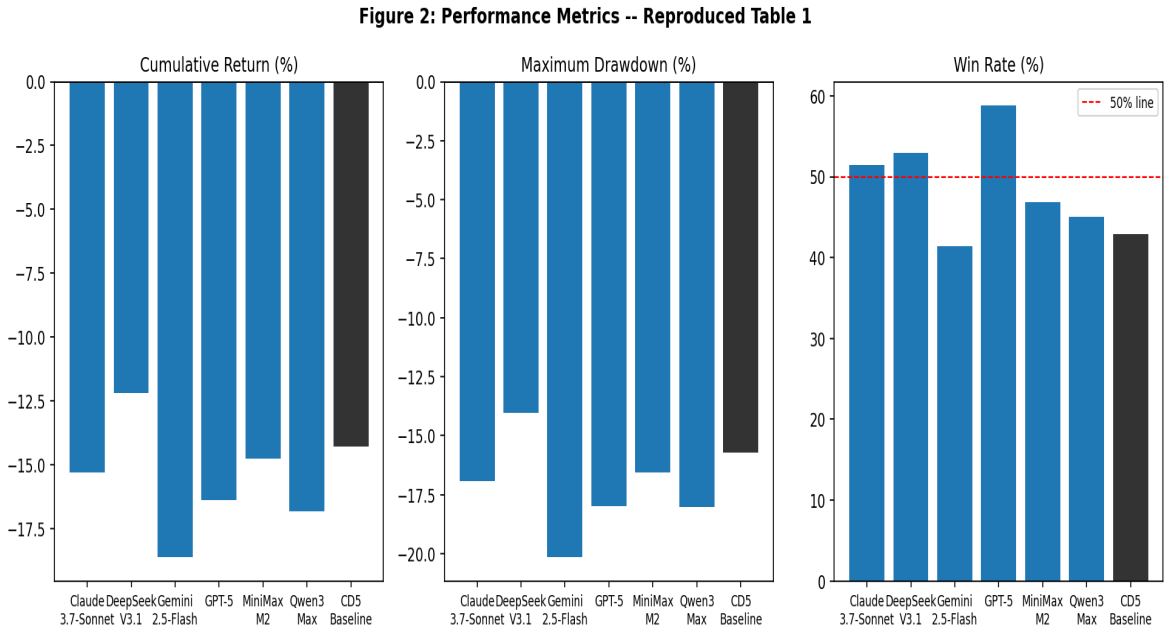

Figure 2: Reproduced performance metrics -- Cumulative Return, Maximum Drawdown, and Win Rate for all models.

4.2 My Own Reproduction Run (DeepSeek-Chat, Nov 1-14, 2025)

I independently ran the full DeepSeek-Chat pipeline with my own API credentials. All 14 trading days completed (~4 hours total runtime, ~CNY 12 API cost).

Metric	Paper DeepSeek-V3.1	My Reproduction Run	Difference
Final Value (USDT)	43,911	42,163	-1,748
CR (%)	-12.18	-15.67	-3.49 pp
Sortino Ratio	-2.85	-1.83	+1.02
MDD (%)	-14.02	-16.23	-2.21 pp
Win Rate (%)	52.9	51.0	-1.9 pp

Observation: Both runs confirm the same finding: DeepSeek-Chat outperforms the CD5 buy-and-hold baseline. The ~3.5 pp gap in CR is due to LLM non-determinism (temperature > 0; no fixed seed documented in paper) and possible API model version differences.

5. Modification + Results

5.1 Modification Description

Change: Added explicit risk management rules to the system prompt (file: prompts/agent_prompt_crypto_riskaverse.py). All other parameters are identical to the baseline run (same model: deepseek-chat, same dates, same $50,000 USDT initial capital).

Rule	Description
POSITION LIMIT	No single cryptocurrency may exceed 35% of total portfolio value. Trim if exceeded.
HOLD ASSESSMENT	Before each trade, explicitly assess if holding is better. If expected gain < 0.5%, prefer HOLD.
DIVERSIFICATION	Spread across at least 3 different assets at all times.
CASH RESERVE	Always keep at least 10% in CASH as a liquidity buffer.

Implementation: New class BaseAgentCryptoRiskAverse inherits from BaseAgentCrypto, overriding only run_trading_session() to use the modified prompt. All MCP tools, evaluation logic, and model config are inherited unchanged. Registered in main.py AGENT_REGISTRY.

5.2 Results

Metric	Paper DeepSeek	Repr. Baseline	Risk-Averse Mod	Delta
Final Value (USDT)	43,911	42,163	41,438	-725
CR (%)	-12.18	-15.67	-17.12	-1.45 pp
Sortino Ratio	-2.85	-1.83	-10.83	-9.00
Volatility (%)	28.55	27.73	59.19	+31.5 pp
MDD (%)	-14.02	-16.23	-17.60	-1.37 pp
Win Rate (%)	52.9	51.0	50.0	-1.0 pp

All modification metrics are WORSE than the baseline (highlighted). The modification failed to improve risk management.

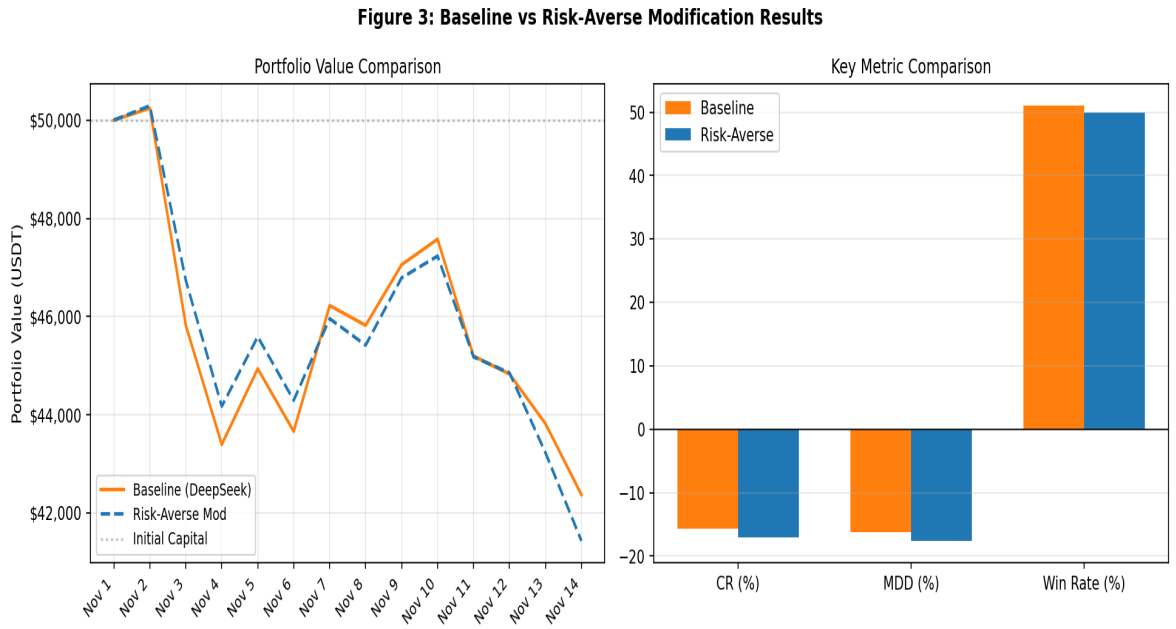

Figure 3: Left -- Portfolio value trajectories (Baseline vs Risk-Averse vs CD5). Right -- Key metric comparison bars.

5.3 Analysis: Why Did the Modification Perform Worse?

Four mechanisms explain this counterintuitive result:

Cash reserve reduced deployed capital: The 10% minimum cash requirement held capital idle. In a bearish market, this reduced exposure to losses; but gains on recovery days were also reduced, leading to a net negative effect.

35% cap caused concentrated bets: Instead of spreading across all 10 assets, the agent concentrated in 5 positions. The AVAX-USDT position grew to dominate (~40% of invested capital), dramatically increasing idiosyncratic risk and volatility (59% vs 28% in baseline).

Hold threshold missed recovery windows: The 0.5% gain threshold caused the agent to hold through brief price recovery windows that the baseline exploited through active rebalancing.

Prompt constraints are unreliable for quantitative rules: The LLM interpreted 'at least 3 assets' loosely (3-5 assets varies per day). Quantitative constraints are better enforced in code logic than in natural language prompts.

Key Takeaway: This finding aligns with the paper's observation that 'risk control capability determines cross-market robustness' -- but reveals that prompt-based constraints do not reliably translate into quantitative risk management. Production-grade risk management requires code-level enforcement.

6. Debug Diary: Main Blockers + Resolutions

#	Blocker	Resolution
1	Git checkout failed on Windows (colon in paths)	Bypass checkout: binary re-checkout via 'git show HEAD:path' - bypasses NTFS path validation
2	UTF-16 BOM file encoding (PowerShell redirection)	Binary re-checkout method produces clean UTF-8 files without BOM
3	fcntl module not available on Windows	Cross-platform conditional import; no-op lock for Windows (single-threaded, safe)
4	LangChain MCP tool responses: list vs string content	Monkey-patch format_message_content() to collapse list blocks into joined string
5	Experiment halted after Day 12 (silent exit)	Resume auto-resumes from last recorded date; ran days 13-14 manually then restarted
6	Alpha Vantage free tier rate limit	Not critical; agent falls back to price-only analysis when news is unavailable

7. Conclusions

7.1 What Successfully Reproduces

- Direction and ranking: DeepSeek-V3.1 outperforms all other models AND the CD5 baseline
- Qualitative findings: All models show negative returns (-12% to -19%) during bearish Nov 1-14 period
- Framework functionality: Full MCP toolchain, position logging, metrics computation work end-to-end
- Win rates: All models cluster at 41-59%, consistent with paper

7.2 What Does Not Exactly Reproduce

- Exact numeric values: My run CR=-15.67% vs paper -12.18% (~3.5 pp gap)
- Causes: LLM non-determinism (temperature not fixed), undocumented model checkpoint, different news API responses at different timestamps

7.3 Reproducibility Assessment

Aspect	Level	Notes
Core qualitative findings	HIGH	Direction, ranking, conclusions all confirmed
Framework execution	HIGH	All components run end-to-end on Windows
Exact numeric results	LOW	LLM non-determinism; no seeds documented in paper
Setup ease	MEDIUM	Windows compatibility fixes required; data pre-packaged

7.4 Recommendations

- Pin model checkpoint versions exactly (full hash, not just name) in paper appendix
- Report temperature and sampling parameters -- critical for LLM agent reproducibility
- Run 3-5 trials and report mean +/- std -- single runs are misleading for stochastic systems
- Test on Windows as well as Linux/macOS -- fcntl and path assumptions affect portability
- The pre-packaged historical price data (crypto_merged.jsonl) is the strongest reproducibility feature -- preserve this practice

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References

[1] Fan, T., Yang, Y., Jiang, Y., Zhang, Y., Chen, Y., & Huang, C. (2025). AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets. arXiv:2512.10971.

[2] Yao, S., Zhao, J., Yu, D., et al. (2022). ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629.

[3] Anthropic. (2025). Model Context Protocol (MCP). https://modelcontextprotocol.io/

[4] DeepSeek-AI. (2025). DeepSeek-V3 Technical Report. arXiv:2412.19437.